AI-DRIVEN PERSONALIZED MARKETING INTELLIGENCE

An Individual Component of the Smart Omni-Retail Recommendation System

Wijetunga J A R K

IT22146588

Bachelor of Science (Hons) in Information Technology
Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

October 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the non-exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Name: Wijetunga J A R K Student ID: IT22146588

Signature: _____ Date: _____

I have also disclosed all use of generative AI and other automated tools in accordance with the module policy (Appendix B), and I remain responsible for the accuracy, originality, citations, code, data, analyses and conclusions in this dissertation.

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Role and name	**Signature**	**Date**
Supervisor: Dr. Dinuka Wijendra		
Co-supervisor: Ms Jenny Kishara		

Abstract

Mid-sized fashion retailers in Sri Lanka send promotions through e-mail, SMS, app notifications and in-store displays, but usually choose the offer, audience and creative by judgement, without knowing which offer a particular customer is most likely to redeem. This dissertation reports the design, implementation and evaluation of AI-Driven Personalized Marketing Intelligence, one of four components of the Smart Omni-Retail System. The component combines a cleaning pipeline that removes personal identifiers; customer intelligence (segments, a recency-frequency-monetary purchase-intent score and disengagement risk); a Random Forest model that estimates the probability that a customer will redeem each of six offer types and ranks them; a poster generator in which the Gemini image model produces only the artwork while dates, venue, terms and the brand logo are drawn deterministically; e-mail delivery and scheduling; a promotion calendar; a key-performance-indicator dashboard with PDF reports; and a read-only link to overstock data from the group's inventory component. The model was trained on a synthetic dataset of 29,996 campaign records for 2,500 customers (2018–2025). A leakage audit found that the supplied past-redemption feature contained the outcomes of the current and later campaigns, which had inflated earlier test results to as much as 85.5% accuracy. After the feature was recomputed from earlier campaigns only, a tuned Random Forest reached 63.3% accuracy, a ROC-AUC of 0.672 and an F1 score of 0.711 (five-fold ROC-AUC 0.673 ± 0.007; 2025 hold-out 0.666), against a 55.9% majority baseline. Campaigns whose offer matched the model's top-ranked offer were redeemed at 61.6%, against 54.7% otherwise. Twenty-two of twenty-five functional tests passed and three defects were identified. Because the data is synthetic and user validation is pending, the results show technical feasibility rather than commercial impact.

Keywords: personalised marketing, promotion recommendation, random forest, data leakage, generative AI

ACKNOWLEDGEMENTS

I would like to thank my supervisor, **Dr. Dinuka Wijendra**, and my co-supervisor, **Ms Jenny Kishara**, for their constant guidance, feedback and time throughout this research.

I am grateful to the PP1 (Progress Presentation One) and PP2 (Progress Presentation Two) evaluation panels for their comments, which shaped the direction of this work. I also thank my group members, Methsara W.M, T A Silva and L A D A S Perera, for the collaboration that made the Smart Omni-Retail Recommendation System possible, in particular for making the inventory component's overstock data available to the marketing component, and the staff of the Faculty of Computing at the Sri Lanka Institute of Information Technology.

I am also grateful to the retail and marketing professionals whose informal conversations and publicly shared insights on LinkedIn helped shape my understanding of how promotional campaigns are planned in practice, and for their insights on customer dissatisfaction and other stakeholders' points of view.

No external funding or sponsorship was received for this research. The dataset was generated by the author, and the use of generative AI tools is disclosed in Appendix B.

TABLE OF CONTENTS

DECLARATION i
Abstract ii
ACKNOWLEDGEMENTS iii
LIST OF FIGURES vi
LIST OF TABLES vii
LIST OF ABBREVIATIONS viii
LIST OF APPENDICES ix
1 INTRODUCTION 1
1.1 Background and Literature Survey 1
1.1.1 Overall project context 1
1.1.2 Individual component context 2
1.1.3 Literature survey method 2
1.1.4 Marketing analytics and AI in marketing 3
1.1.5 Customer value, engagement and RFM 3
1.1.6 Response modelling and uplift modelling 3
1.1.7 Tree ensembles for tabular data 4
1.1.8 Evaluation of probabilistic classifiers and data leakage 4
1.1.9 Generative image models 5
1.1.10 Existing systems, products and techniques 5
1.1.11 Summary of the literature survey 6
1.2 Research Gap 7
1.3 Research Problem 8
1.4 Research Objectives 9
1.4.1 Main objective 9
1.4.2 Specific objectives 9
1.4.3 Research questions 10
1.4.4 Scope, limitations and assumptions 10
1.4.5 Structure of the dissertation 11
2 METHODOLOGY 12
2.1 Methodology 12
2.1.1 Research design 12
2.1.2 Data and study materials 13
2.1.3 Features and target 14
2.1.4 Leakage audit of the past-redemption feature 15
2.1.5 Model selection and training 16
2.1.6 Evaluation measures and data analysis 16
2.1.7 Requirements 17
2.1.8 System architecture and design decisions 18
2.1.9 Upload, cleaning and storage design 19
2.1.10 Redemption model and offer recommender design 20
2.1.11 Customer intelligence design 20
2.1.12 Poster generation design 21
2.1.13 Delivery, scheduling and calendar design 21
2.1.14 KPI dashboard, reports and inventory link design 22
2.1.15 Ethics, privacy, security and responsible AI 22

2.2 Commercialization Aspects of the Product 23
2.2.1 Value proposition and target users 23
2.2.2 Market and competitive position 23
2.2.3 Business and sustainability model 23
2.2.4 Scalability, intellectual property and compliance 23
2.2.5 Contribution to the Sustainable Development Goals 24
2.3 Testing and Implementation 24
2.3.1 Development environment and technology stack 24
2.3.2 Component implementation 25
2.3.3 Integration with the other components 26
2.3.4 Verification and software testing 27
2.3.5 Deployment, standards and reproducibility 28
3 RESULTS AND DISCUSSION 29
3.1 Results 29
3.1.1 Redemption model performance 29
3.1.2 Effect of the leakage fix 30
3.1.3 Calibration and targeting value 32
3.1.4 Offer ranking 33
3.1.5 Product, integration and user validation results 34
3.2 Research Findings 35
3.3 Discussion 35
3.3.1 Interpretation of findings 35
3.3.2 Comparison with literature and existing solutions 36
3.3.3 Limitations and threats to validity 37
3.3.4 Implications 37
3.4 Summary of Each Student's Contribution 38
4 CONCLUSIONS AND RECOMMENDATIONS 39
4.1 Conclusions 39
4.2 Achievement of Objectives 39
4.3 Recommendations and Future Work 40
REFERENCE LIST 41
GLOSSARY 44
APPENDICES 45
Appendix A: Extended Results and Research Instruments 45
A.1 Leakage fix in the training script 45
A.2 Full test-case record 45
A.3 Model and environment details 46
A.4 User validation evidence and instruments 47
Appendix B: Academic Integrity and AI Use Disclosure 47
B.1 Declaration statement 48

LIST OF FIGURES

Figure 1.1: Overall Smart Omni-Retail Recommendation System context 1
Figure 1.2: Conceptual framework of the marketing intelligence component 8
Figure 2.1: Research methodology workflow and evolution of the component 12
Figure 2.2: Redemption rate by offer type and by timing context 14
Figure 2.3: Supplied and corrected past-redemption rate for one customer (CUST_1577) 15
Figure 2.4: Architecture of the marketing intelligence component 18
Figure 2.5: Firestore collections used by the marketing component 20
Figure 2.6: Poster generation pipeline 21
Figure 2.7: Lifecycle of a scheduled poster 22
Figure 2.8: Integration with the inventory component 27
Figure 3.1: Test ROC-AUC with the supplied (leaking) and corrected feature 31
Figure 3.2: Feature importance of the selected random forest 32
Figure 3.3: Calibration of predicted redemption probabilities (test set, deciles) 33
Figure 3.4: Recommender screen with ranked offers and inventory notes 34
Figure 3.5: Example generated poster with deterministic event band and logo 34
Figure 3.6: Hub dashboard, customer intelligence and calendar 34

LIST OF TABLES

Table 1.1: Components of the Smart Omni-Recommendation Retail System..... 2
Table 1.2: Indicative comparison of approaches..... 6
Table 1.3: Summary of the literature survey..... 7
Table 1.4: Specific objectives..... 9
Table 1.5: Research questions..... 10
Table 2.1: Attributes of the campaign dataset..... 13
Table 2.2: Summary of the campaign dataset..... 14
Table 2.3: Models compared..... 16
Table 2.4: Functional requirements..... 17
Table 2.5: Non-functional requirements..... 18
Table 2.6: Key design decisions and alternatives..... 19
Table 2.7: Relationship to the Sustainable Development Goals..... 24
Table 2.8: Technology stack..... 25
Table 2.9: Backend modules and endpoints..... 26
Table 2.10: Defects found and corrective actions..... 28
Table 3.1: Model performance on the held-out test set (corrected feature, n = 6,000) 29
Table 3.2: Robustness of the selected random forest..... 30
Table 3.3: Test ROC-AUC (accuracy) with the supplied and corrected feature..... 30
Table 3.4: Feature-group ablation of the selected random forest (test set)..... 31
Table 3.5: Example recommendation for one test customer..... 34
Table 3.6: Research contributions and strength of evidence..... 35
Table 3.7: Limitations and their effects..... 37
Table 3.8: Summary of each student's contribution..... 38
Table 4.1: Objective-achievement matrix..... 40
Table 4.2: Recommendations and future work..... 40
Table A.1: Detailed test-case record..... 45
Table A.2: Model configuration and environment..... 47
Table A.3: User validation evidence to be added..... 47
Table B.1: AI use disclosure..... 48

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
BOGO	Buy One Get One
CSV	Comma-Separated Values
FR	Functional Requirement
KPI	Key Performance Indicator
LKR	Sri Lankan Rupee
NFR	Non-Functional Requirement
PDF	Portable Document Format
PDPA	Personal Data Protection Act
PII	Personally Identifiable Information
PP1, PP2, PP3	Progress Presentations 1, 2 and 3
REST	Representational State Transfer
RFM	Recency, Frequency, Monetary value
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
SMS	Short Message Service
SMTP	Simple Mail Transfer Protocol
TLS	Transport Layer Security
XGBoost	Extreme Gradient Boosting

LIST OF APPENDICES

Appendix	Description	Page
Appendix A	Extended Results and Research Instruments	45
Appendix B	Academic Integrity and AI Use Disclosure	47

1 INTRODUCTION

1.1 Background and Literature Survey

1.1.1 Overall project context

Fashion retail in Sri Lanka is increasingly multi-channel. A mid-sized retailer typically runs several showrooms, keeps a loyalty programme and reaches customers through e-mail, short message service (SMS), a mobile app and in-store displays, but rarely has the analytics budget of a global chain. The group research project, the Smart Omni-Recommendation Retail System, was developed for a retail setting inspired by a Sri Lankan multi-branch fashion retailer. No confidential company records were used; every dataset in this dissertation was generated synthetically.

The system has four integrated components that use artificial intelligence (AI), each developed by a different group member, as shown in Figure 1.1 and Table 1.1. My component, AI-Driven Personalized Marketing Intelligence, turns customer and campaign data into recommended offers, campaign materials, and performance reports, and consumes overstock signals from the inventory component so that promotions can help clear surplus stock.

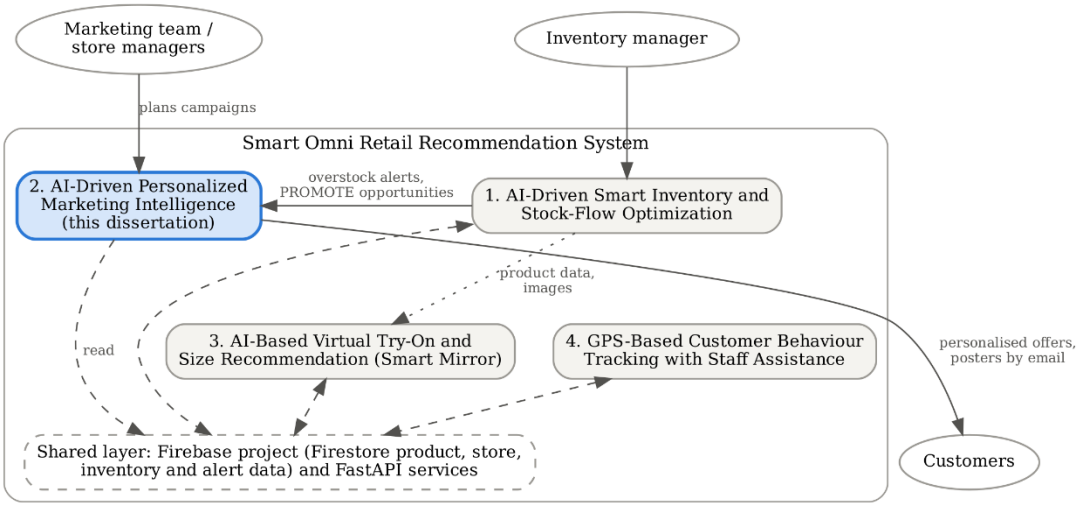


Figure 1.1: Overall Smart Omni-Retail Recommendation System context

Table 1.1: Components of the Smart Omni-Retail Recommendation System

No.	Component	Member	Student ID
1	AI-Driven Smart Inventory and Stock-Flow Optimization	Methsara W.M	IT22244598
2	AI-Driven Personalized Marketing Intelligence	Wijetunga J A R K	IT22146588
3	AI-Based Virtual Try-On and Size Recommendation (Smart Mirror)	T A Silva	IT22284952
4	GPS-Based Customer Behaviour Tracking with Staff Assistance	L A D A S Perera	IT22243812

1.1.2 Individual component context

The component supports the people who plan and run promotions: a marketing executive at head office and store managers who promote events in their branches. Instead of handling customer analysis, offer selection, creative design, delivery and measurement in separate tools, it brings them together in one Flutter application with seven tabs. It accepts and cleans campaign, customer and transaction files, removing personally identifiable information (PII) before storage; describes the customer base through segments, a purchase-intent score and disengagement risk; estimates the probability that a customer will redeem each of six offer types and ranks them; generates posters with a generative image model while placing all factual text deterministically; sends or schedules posters by e-mail with calendar reminders; reports campaign performance on a dashboard and in Portable Document Format (PDF) reports; and reads promotion opportunities flagged by the inventory component, which can be turned into a poster in one action.

1.1.3 Literature survey method

The survey combined a review of published literature with informal domain research. The literature review covers marketing analytics and AI in marketing, customer value measures, response and uplift modelling, tree ensembles for tabular data, evaluation and data leakage, and generative image models. Sources were identified mainly through Google Scholar and IEEE Xplore, supplemented by openly available sources such as open-access journal and conference papers and official technical documentation. The search was targeted rather than systematic, prioritising peer-reviewed work from 2016 to 2026 and older sources where they define a foundational method.

To ground the work in practice, the literature was complemented by informal conversations with members of a marketing department and with owners of small retail businesses. These conversations explored how promotions are planned and approved, which customer details are needed to target a promotion (such as purchase history, visit frequency, loyalty tier and preferred categories), and how stakeholders such as marketing, store and inventory staff work together when running a campaign. These insights informed the choice of customer attributes in the synthetic SkyHigh dataset and the features of the component. They were exploratory discussions rather than a formal study, so no personal data was collected and no findings from them are reported as research results.

1.1.4 Marketing analytics and AI in marketing

Wedel and Kannan describe the shift from aggregate, intuition-led planning to analytics at the level of the individual customer, with personalisation and campaign optimisation as central applications [1]. Davenport et al. place offer personalisation and content creation among the most immediate uses of AI in marketing [2], and Huang and Rust propose combining mechanical, thinking and feeling AI across marketing research, strategy and action [3]. The component follows that pattern: rules and analytics describe customers, a learned model supports the offer decision, and a generative model supports content, with the marketer keeping the final decision. Promotions are not interchangeable: Chandon, Wansink and Laurent show that

monetary and non-monetary promotions deliver different benefits whose effectiveness depends on the customer [4], which supports predicting the best offer type per customer.

1.1.5 Customer value, engagement and RFM

Recency, frequency and monetary value (RFM) are long-standing summaries of customer behaviour; Fader, Hardie and Lee show that they are sufficient statistics for a family of customer-lifetime-value models [5]. Retailers often combine normalised RFM scores into one weighted score, which is the heuristic used for the purchase-intent score in this work. For disengagement, Verbeke et al. argue that churn models should be judged by the profit they enable [6], and Ascarza shows that the customers at highest risk are not necessarily those who respond best to retention offers [7]. The at-risk measure is therefore reported for monitoring, and targeting is left to the redemption model and the marketer.

1.1.6 Response modelling and uplift modelling

A response model estimates the probability that a customer responds given their attributes and the campaign. An uplift model estimates the change in that probability caused by the campaign, by comparing treated and untreated customers [8], [9]; Künzel et al. describe meta-learners such as the T-learner for this purpose [10]. A customer who would buy anyway has a high response probability but no uplift. The recommender in this work is a response model conditioned on offer type: it ranks offers by the probability of redemption, which helps choose between offers but is not a causal estimate. The dashboard reports the treatment-versus-control difference at campaign level, and Section 4.2 proposes uplift modelling as future work. Rzepakowski and Jaroszewicz extend decision trees to multiple treatments, which corresponds to choosing among several offers [9], but such studies evaluate models offline and seldom describe a tool in which a marketer can explore how the predicted outcome changes with offer, discount or timing.

1.1.7 Tree ensembles for tabular data

Customer and campaign records are tabular data with mixed categorical and numeric attributes. A benchmark across 45 tabular datasets found that tree-based ensembles remained state of the art on medium-sized tabular data [11]. A random forest averages many decision trees, each grown on a bootstrap sample with a random subset of features at each split, which reduces variance and makes the model robust to noisy features [12]; its depth and minimum leaf size control how closely it can fit individual records. Gradient boosting, implemented efficiently in Extreme Gradient Boosting (XGBoost) [13], instead fits trees sequentially to the errors of earlier trees, and logistic regression [14] is the standard linear reference. All three were compared in this work. Feature-attribution methods such as SHapley Additive exPlanations (SHAP) can explain individual predictions of tree ensembles [15], and explainability is increasingly expected of AI used in decisions about people [16].

1.1.8 Evaluation of probabilistic classifiers and data leakage

The area under the receiver operating characteristic curve (ROC-AUC) measures how well scores rank positive cases above negative ones, independently of a threshold [17], and is the most relevant measure when scores are used to rank options. Because the recommender shows probabilities, calibration also matters; tree ensembles can be poorly calibrated without correction [18]. Kaufman et al. define leakage as the use of information about the target that would not be available when the model is used [19], and Kapoor and Narayanan found leakage in many published machine-learning studies, often through features computed with information from the test set or the future [20]. A customer's past redemption rate computed over all of the customer's records is a classic example.

1.1.9 Generative image models

Diffusion models generate images by learning to reverse a noising process [21]; latent diffusion made high-resolution text-to-image generation practical [22], and systems such as DALL-E 2 [23] and Imagen [24] produce photorealistic images from prompts. These models render text poorly: generated text is often misspelled or malformed [24], [25]. Multimodal models such as Google's Gemini image models generate images

from a conversational prompt through an application programming interface (API) [26]. Multimodal models such as Google's Gemini image models generate images from a conversational prompt through an application programming interface (API) [26]. This component uses Gemini 2.5 Flash Image (model identifier gemini-2.5-flash-image, also known as "Nano Banana"), chosen because its free API tier is sufficient for the scale of an academic prototype. Google has since released newer image models in the same series, so the model identifier is kept in one place in the backend to allow a later upgrade. While AI-generated marketing content is widely anticipated [2], [3], this unreliability means that a poster's facts cannot be left to the model.

1.1.10 Existing systems, products and techniques

Practice falls into four broad approaches. The first is manual planning with spreadsheets and general design tools, where the offer and audience are chosen by experience. The second is e-mail marketing platforms such as Mailchimp, which provide audience segmentation based on customer data, scheduled sending and campaign reports, and on higher plans predictive segments that estimate customer value [31]. The third is customer-relationship and marketing-automation platforms with built-in analytics. For example, Klaviyo predicts each customer's lifetime value, churn risk and expected date of next order [32], and Salesforce Marketing Cloud's Einstein Engagement Scoring predicts how likely each subscriber is to open an e-mail, click a link, stay subscribed and convert [33]. The fourth is design tools with generative-AI features, such as Canva's AI image generator, which turns a text prompt into an image [34]. Based on their public documentation, these tools score customers or engagement but do not rank several offer types for the same customer, and none links promotion planning to inventory surplus. Table 1.2 is my own indicative synthesis.

Table 1.2: Indicative comparison of approaches

Table 1.2: Indicative comparison of approaches

Approach	Offer choice per customer	Creative production	Delivery and measurement	Link to inventory
Manual / spreadsheet	Judgement	Manual	Manual, partial	Informal
E-mail and SMS platforms	Rule-based segments	Templates	Strong	Usually none
Customer-relationship and loyalty analytics	Propensity scores in some products	Limited	Varies	Varies
Generative design tools	None	Strong visuals, unreliable text	None	None
This component	Ranks six offer types by predicted redemption	Generated artwork with deterministic facts	E-mail, scheduling, key performance indicators (KPIs), PDF reports	Reads overstock and PROMOTE opportunities

1.1.11 Summary of the literature survey

Table 1.3 summarises what the reviewed literature establishes in each area and what it leaves open for the problem addressed in this dissertation.

Table 1.3: Summary of the literature survey

Area	Key sources	What the literature establishes	What remains open for this problem
Marketing analytics and AI	[1], [2], [3]	Customer-level analytics and AI improve targeting and content creation	Few integrated tools that a small marketing team can use end to end
Promotion types	[4]	Monetary and non-monetary promotions work differently for different customers	Choosing the offer type per customer from data
Customer value and engagement	[5], [6], [7]	RFM summarises value; risk scores do not identify who responds best	Combining engagement monitoring with offer choice
Response and uplift modelling	[8], [10], [9]	Uplift models estimate incremental effects, including multiple treatments	Interactive comparison of offers and timing for one customer
Tree ensembles	[12], [13], [11]	Tree ensembles are state of the art on medium-sized tabular data	Choosing and regularising a model on customer-history data
Evaluation and leakage	[17], [18], [19], [20]	ROC-AUC and calibration suit ranking; leakage inflates results	Leakage checks for customer-history features in applied tools
Generative image models	[21], [22], [23], [24], [25]	High-quality images from text prompts, but unreliable text rendering	Keeping the facts on a marketing poster exact

1.2 Research Gap

Data-driven targeting is known to improve campaign response [1], [2], [3], and established techniques exist for each part of the problem: RFM analysis [5], churn and retention modelling [6], [7], response and uplift modelling [8], [10], [9], tree ensembles for tabular prediction [12], [11], and text-to-image generation [22], [23], [24]. In the sources reviewed, however, these are studied separately. Response models usually score a single, already-chosen offer rather than compare several offers for one customer; studies of image generation report weak text rendering but do not address how a retail poster can be kept factually correct; and stock data is outside the scope of marketing studies. A methodological gap concerns evaluation: customer-history features such as a past redemption rate are especially exposed to leakage [19], [20], because the convenient definition includes the campaign being predicted, yet applied projects often report accuracy without checking for it.

The gap addressed here is therefore the lack of an integrated, marketer-facing component that (a) ranks several offer types for one customer with a leakage-checked redemption model, (b) uses generative AI for artwork while keeping every stated fact deterministic, (c) closes the loop from delivery to measurement, and (d) links promotion planning to inventory surplus. This claim is limited to the sources reviewed. Figure 1.2 shows the conceptual framework that follows from it: campaign history, customer profiles, timing context and inventory signals form the data layer; the component describes customers, scores every offer, creates posters with generated artwork and deterministic facts, delivers campaigns and measures their results, which feed the next cycle, with the marketer in control at the decision, creation and delivery stages.

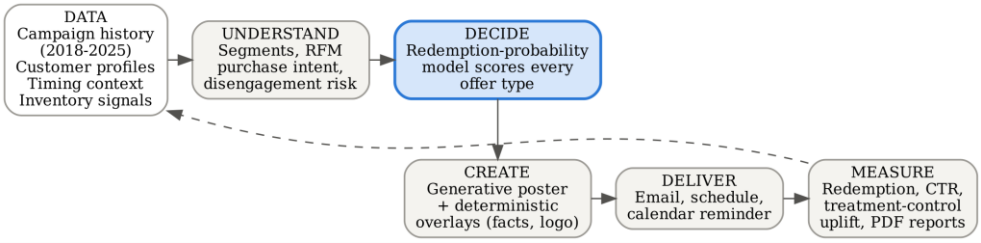


Figure 1.2: Conceptual framework of the marketing intelligence component

1.3 Research Problem

Every discount reduces margin, and every message sent to a customer who ignores it adds cost. Mid-sized retailers nevertheless tend to send the same offer to large groups of customers, chosen by intuition. Three difficulties make this hard to improve. First, the marketer cannot see which of several offer types (a flash sale, a buy-one-get-one (BOGO) deal, a loyalty reward and so on) a given customer is most likely to redeem, or how much campaign timing matters. Second, producing creative takes time and design skill; generative image models can help but render exact text such as dates, prices and venues unreliably [24], [25], and that is the information a promotion poster must get right. Third, campaign results are seldom fed back systematically, and promotions are planned without reference to stock, even though clearing overstock is a main reason to run them.

The research problem is therefore: how can a mid-sized multi-channel fashion retailer be given a single, privacy-conscious tool that reliably estimates which offer each customer is most likely to redeem, helps produce accurate campaign material quickly, delivers and measures campaigns, and connects promotion planning to inventory surplus?

1.4 Research Objectives

1.4.1 Main objective

The main objective is to design, implement and evaluate an integrated AI-driven marketing intelligence component for a multi-channel fashion retailer that recommends personalised offers, generates and delivers accurate campaign material, measures campaign performance and uses inventory signals from the wider system.

1.4.2 Specific objectives

Table 1.4 lists the specific objectives; Section 4.1 reports how far each was achieved.

Table 1.4: Specific objectives

ID	Objective	Addressed in
O1	Prepare a multi-year campaign dataset and an upload pipeline that cleans data and removes personal identifiers before storage	Sections 2.1.2, 2.3.2
O2	Develop and evaluate a leakage-free model of redemption probability against simple baselines	Sections 2.1.4, 3.1
O3	Provide a recommender that scores every offer type for a customer profile and timing context	Sections 2.1.10, 3.1.4
O4	Provide customer intelligence: segments, an RFM purchase-intent score and disengagement risk	Sections 2.1.11, 3.1.5
O5	Implement poster generation in which all factual content is rendered deterministically	Section 2.1.12
O6	Implement campaign delivery by e-mail, scheduling and a promotion calendar	Section 2.1.13
O7	Provide a KPI dashboard and PDF reports that share one calculation	Sections 2.1.14, 3.1.5
O8	Integrate read-only with the inventory component and verify the component by testing	Sections 2.3.3, 2.3.4

1.4.3 Research questions

Four research questions guide the work (Table 1.5). They are framed as questions because the evaluation uses classification metrics, cross-validation and functional verification rather than formal hypothesis tests.

Table 1.5: Research questions

ID	Research question	Evidence
RQ1	How well can a tree-ensemble model estimate a customer's probability of redeeming a given offer, compared with simple baselines, once leakage is removed?	Section 3.1
RQ2	Does ranking all offer types by predicted probability identify offers that are redeemed more often than the offers actually sent?	Section 3.1.4
RQ3	Can generative image AI be used for posters while keeping dates, venues, terms and branding exactly as specified?	Sections 2.1.12, 2.3.4
RQ4	Can inventory surplus signals from another component be used in campaign planning without risk to that component's data?	Section 2.3.3

1.4.4 Scope, limitations and assumptions

The component covers data upload and cleaning, customer intelligence, the redemption recommender, poster generation, e-mail delivery and scheduling, the calendar, the dashboard and reports, and the read-only inventory link, delivered as a Flutter application with a FastAPI backend and Firebase Firestore storage. All data is synthetic, so model results describe how well the patterns built into the generator are recovered, not real-world accuracy. The recommender estimates redemption probability, not the causal effect of sending an offer (Section 1.1.6). There is no live link to point-of-sale or customer-relationship systems, poster generation and e-mail depend on external services, scheduled posts are sent manually, and no monetary benefit was measured. Customer attributes are assumed fixed per customer, as in the data; timing flags are assumed known when a campaign is planned; and the marketer reviews every poster before it is sent.

1.4.5 Structure of the dissertation

Chapter 2 describes the methodology. Section 2.1 explains the research design and how the component evolved, describes the synthetic campaign dataset, sets out the features and target of the redemption model, reports the leakage audit of the past-redemption feature, and explains how the models were selected, trained and evaluated. It then presents the requirements, the system architecture and design decisions, and the detailed design of each feature: upload and cleaning, the offer recommender, customer intelligence, poster generation, delivery and scheduling, and the dashboard, reports and inventory link, followed by the ethical, privacy and security considerations. Section 2.2 discusses the commercialisation aspects of the product, including its value proposition, market position, business model, compliance and contribution to the Sustainable Development Goals. Section 2.3 covers the technology stack, the implementation, the integration with the inventory component, the verification tests and the deployment.

Chapter 3 presents the results of the model evaluation, the effect of the leakage fix, the calibration and offer-ranking analyses, and the product and integration results. It then states the research findings, discusses them against the literature and existing solutions, sets out the limitations and implications, and summarises each student's contribution to the group project.

Chapter 4 draws the conclusions, answers the research questions, reviews the achievement of each objective and makes recommendations for future work. The reference list, a glossary of key terms and the appendices follow, including the full test-case record, the model configuration and the disclosure of generative AI use.

2 METHODOLOGY

2.1 Methodology

2.1.1 Research design

The study follows a design-and-evaluate approach (Figure 2.1). The redemption model was assessed quantitatively with baselines, five-fold cross-validation and a chronological test; the other features were assessed by scenario-based testing of the backend API.

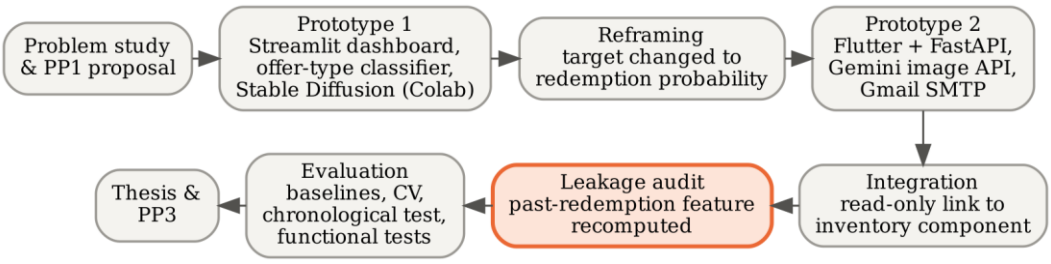


Figure 2.1: Research methodology workflow and evolution of the component

The project changed direction twice. The first prototype was a Streamlit dashboard with a T-learner uplift model, an XGBoost model that predicted which offer type had been sent to a customer, and a Stable Diffusion poster generator reached through Colab and an ngrok tunnel. The offer-type target could not be predicted well, because offers had been assigned close to randomly for treatment-control analysis; the usable signal is whether a customer redeems the offer received. The model was therefore reframed to predict redemption probability with the offer type as an input. The second prototype moved to Flutter and FastAPI, replaced Stable Diffusion with the Gemini image API, and added e-mail, scheduling, the calendar, reports and the inventory link. The second change came from the evaluation itself: when the model was re-examined for this dissertation, its earlier accuracy was found to depend on a leaking feature (Section 2.1.4), and the final model was re-selected on corrected data.

2.1.2 Data and study materials

The main dataset was generated synthetically by the author to represent the campaign history of a Sri Lankan fashion retailer. It contains 29,996 campaign records for 2,500

customers from 2018 to 2025; each record is one offer sent to one customer, with whether the customer clicked and redeemed it. Campaign volume grows from 2,994 records in 2018 to 4,610 in 2025, and each customer received 3 to 27 campaigns (mean 12). Synthetic data was used because real customer campaign data could not be obtained, and a consistent eight-year history with a treatment indicator was needed. Every result in Chapter 3 therefore measures how well the model recovers patterns built into the generator. This migration followed feedback after PP1 that all four components should share a common architecture so that they could be integrated into one system, and the version demonstrated at PP2 was the integrated Flutter and FastAPI product. Smaller files of 500 customers, 1,000 campaigns, 5,000 transactions and 20 vendors were used to demonstrate upload and analytics. No human participants were recruited for the studies reported here. Table 2.1 lists the attributes and Table 2.2 summarises them.

Table 2.1: Attributes of the campaign dataset

Attribute group	**Fields**
Identifiers	customer_id, campaign_id (synthetic codes; no names or contact details)
Customer profile	gender, age, age_group (18–25, 26–34, 35–45, 46–60), loyalty_tier (Bronze, Silver, Gold, Platinum), preferred_category (10 categories)
Customer behaviour	visit_frequency, total_spend_lkr, avg_basket_value_lkr, days_since_last_purchase, customer_past_redemption_rate
Offer	offer_type (BOGO, Discount, Flash Sale, Loyalty Reward, New Arrival, Seasonal Sale), discount_pct (10–50%), channel (App Push, Email, In-Store, SMS)
Timing	sent_date, year, is_seasonal_window, is_salary_cycle, is_school_holiday
Outcomes and experiment	clicked, redeemed, revenue_after_lkr, treatment (1 = treatment, 0 = control)

Table 2.2: Summary of the campaign dataset

Measure	Value
Records / customers / period	29,996 / 2,500 / 2018–2025
Redemption rate / click rate	55.9% / 66.9%
Offer types / channels	Six (4,926–5,096 records each) / four (7,410–7,561 each)
Treatment / control records	20,824 / 9,172
Redemption, treatment / control	59.2% / 48.5%
Revenue from redeemed campaigns	Sri Lankan rupees (LKR) 178.9 million (mean LKR 10,663 per redemption)

The near-equal counts of offer types and channels are consistent with random assignment. Figure 2.2 shows that timing matters more than offer type: flash sales were redeemed most often (61.7%) and new-arrival offers least (51.3%), but redemption was 63.1% inside seasonal windows against 40.8% outside, and 64.2% during the salary cycle against 50.2% outside.

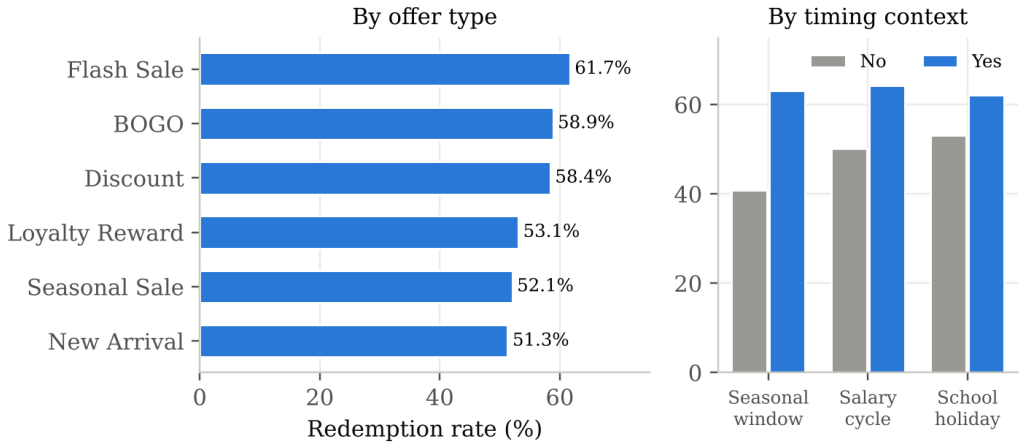


Figure 2.2: Redemption rate by offer type and by timing context

2.1.3 Features and target

The model uses six categorical features (age group, gender, loyalty tier, channel, preferred category, offer type) and nine numeric features (visit frequency, total spend, average basket value, days since last purchase, past redemption rate, discount percentage and three timing flags). Categorical features are label-encoded with the sorted list of observed values, and the encoders are saved with the model so the API

encodes requests identically and rejects unknown values. The target is redeemed. Offer type and discount are inputs: at prediction time the service holds the customer and timing fixed, substitutes each offer type in turn and ranks the six probabilities.

2.1.4 Leakage audit of the past-redemption feature

In the supplied data the past-redemption column behaved suspiciously: almost every customer had exactly two distinct values, and flexible models lost much of their accuracy when it was removed. Comparison with candidate definitions showed that for all 29,996 rows it equalled the customer's redemption rate over all of their other campaigns, including later ones (correlation 1.000), whereas its correlation with a rate from earlier campaigns only was 0.60. This leaks the target twice over. Because the current campaign is excluded, a customer with k redemptions in n campaigns has k/(n−1) on non-redeemed rows and (k−1)/(n−1) on redeemed rows, so within a customer the lower value marks a redemption, which a model can learn because the customer's fixed attributes identify them. And because later campaigns are included, the feature uses future information. Figure 2.3 illustrates this for one customer.

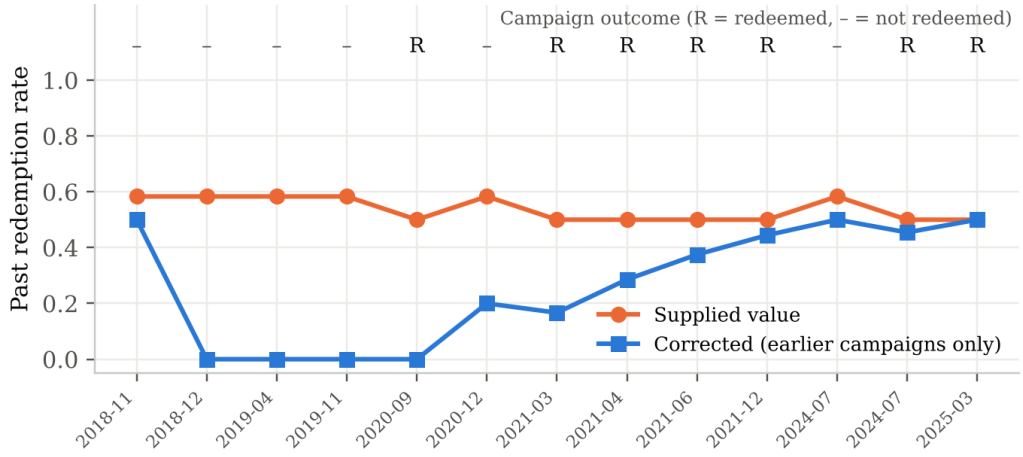


Figure 2.3: Supplied and corrected past-redemption rate for one customer (CUST_1577)

The training script now recomputes the feature from strictly earlier campaigns: records are sorted by customer, sent date and campaign identifier, and earlier redemptions are divided by earlier campaigns; a customer's first campaign receives a neutral 0.5 (2,500 records). A regression test (TC25) confirms that no row uses its own or later outcomes,

and Appendix A.1 lists the code. At prediction time the marketer enters the customer's redemption rate to date, which is the same quantity.

2.1.5 Model selection and training

Table 2.3 lists the models compared. All were trained on the same stratified 80/20 split with random seed 42 (23,996 training and 6,000 test records). The random forest's settings were chosen by a grid search with five-fold cross-validation on the training set only, over maximum depth (8, 10, 12, 16, unlimited), minimum samples per leaf (5, 20, 50) and features per split (square root, half). The best combination, depth 10 with at least 50 records per leaf and the square root of the features at each split, reached a cross-validated ROC-AUC of 0.671; larger leaves consistently helped, because they stop trees from fitting individual customers. The selected model was further checked with five-fold cross-validation on the full dataset and a chronological test that trained on 2018–2024 and tested on the 4,610 campaigns of 2025, which mirrors real use. Every model was trained twice, with the supplied and the corrected feature, to measure the effect of the leak.

Table 2.3: Models compared

Model	Description
Majority class	Predicts the overall training redemption rate for every record
Offer-type historical rate	Predicts the training redemption rate of the record's offer type, a policy a marketer could follow without a model
Logistic regression	One-hot categorical and standardised numeric features [27]
XGBoost	250 trees, depth 5, learning rate 0.08, row and column subsample 0.9 (the earlier production configuration) [13]
Random forest (default)	300 trees, scikit-learn defaults (unlimited depth, leaf size 1) [12]
Random forest (selected)	300 trees, maximum depth 10, minimum 50 samples per leaf, square-root features per split

2.1.6 Evaluation measures and data analysis

ROC-AUC is the primary measure because the recommender ranks offers [17]. Accuracy, precision, recall and F1 are reported at the 0.5 threshold; the Brier score (mean squared error of the probabilities) and a decile calibration plot assess the probabilities shown to users; top-decile lift measures the value of the score for

targeting; and ranking alignment compares the observed redemption rate of test records whose actual offer equalled the model's top-ranked offer with that of the others. Because offers were assigned close to randomly, this is a fair offline indication but not a controlled experiment. Metrics were calculated with scikit-learn [27] on the held-out test set and across folds and are reported to three decimal places with fold standard deviations; functional tests are reported as pass or fail with the observed output.

2.1.7 Requirements

Requirements were derived from the objectives and the needs of two user roles: a marketing executive who plans campaigns and reviews performance, and a store manager who promotes branch events. Table 2.4 lists the functional requirements (FR) and Table 2.5 the non-functional requirements (NFR).

Table 2.4: Functional requirements

ID	Requirement	Objective
FR1	Accept campaign, customer and transaction comma-separated values (CSV) files; clean them, remove personal-identifier columns, store them and report the changes; keep an upload history with download and delete	O1
FR2	Show segments by tier, age and category, new versus returning customers, a purchase-intent score and disengagement risk with an adjustable threshold	O4
FR3	Return, for a customer profile, discount and timing context, the predicted redemption probability of every offer type, ranked, with overstock items in the chosen segment	O2, O3
FR4	Generate a poster from structured options and free text, drawing event details, terms and the brand logo deterministically	O5
FR5	Send a poster by e-mail, or schedule it with a calendar reminder; show a calendar of campaign activity and notes	O6
FR6	Show KPIs for a selectable date range and generate PDF reports for chosen years, months or quarters with the same figures	O7
FR7	List promotion opportunities from the inventory component with filters and create a poster from one	O8

Table 2.5: Non-functional requirements

ID	Quality	Requirement
NFR1	Privacy	Identifier columns removed before storage; recipients never see each other's addresses
NFR2	Correctness	Dashboard and reports share one KPI function; poster facts are never produced by the generative model
NFR3	Isolation	Inventory data accessed through a separate, read-only connection
NFR4	Robustness	Failures of Gemini, the mail server or Firebase return clear errors, never an unhandled crash
NFR5	Cost	Storage stays within Firestore's free tier: no per-row documents, no images in history
NFR6	Security and maintainability	Secrets in environment variables; routes, services and schemas separated; model loaded once and cached

2.1.8 System architecture and design decisions

Figure 2.4 shows the architecture. Each of the seven Flutter tabs (Hub, Upload, Recommender, Poster, Inventory, Calendar, Customers) calls a group of representational state transfer (REST) endpoints on the FastAPI backend, where routes validate requests with Pydantic schemas and delegate to services. Two Firebase connections are kept apart: the component's own project stores uploads, notes, poster history and scheduled posts, and a second, read-only connection reads the shared team project where the inventory component publishes its alerts. Gemini and Gmail are the only external services. Table 2.6 records the main design decisions, the alternatives considered and the reasons for each choice.

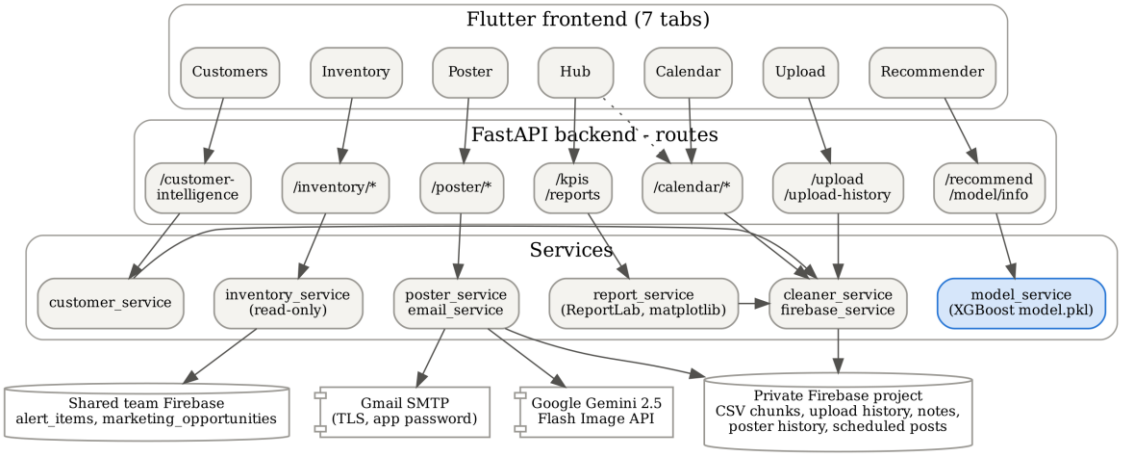


Figure 2.4: Architecture of the marketing intelligence component

Table 2.6: Key design decisions and alternatives

Decision	Alternative considered	Reason for the choice
Predict redemption with offer type as input	Predict which offer was sent	The offer sent was close to random; redemption carries the usable signal and lets every offer be scored
Response model with offer ranking	T-learner uplift (first prototype)	Answers the marketer's question directly and is simpler to serve; uplift kept as a campaign-level KPI
Past-redemption rate from earlier campaigns	Supplied column	The supplied column leaked current and future outcomes (Section 2.1.4)
Regularised random forest	XGBoost, logistic regression	Equal ranking quality within one standard deviation, highest F1 and recall, least affected by the leak, few hyperparameters (Section 3.1)
Gemini image API	Stable Diffusion via Colab and ngrok	No tunnel to keep alive; reliable from a deployed backend
Deterministic overlays for facts	Model writes all text	Generated text is unreliable [24], [25]
Flutter + FastAPI	Streamlit	Separate frontend and API that can be tested and reused directly
CSV chunks of 5,000 lines in Firestore	One document per row	A few writes per upload instead of tens of thousands; within free-tier quotas [28]
History stores configuration only	Store every image	Firestore documents are limited to 1 MiB [28]
Manual send of scheduled posts	Background scheduler	The prototype backend is not always on

2.1.9 Upload, cleaning and storage design

An upload names its type (campaigns, customers or transactions) and is read with a fallback sequence of encodings so that Excel exports are accepted. The cleaning service normalises column names, removes duplicate rows, drops personal-identifier columns, converts dates, strips thousands separators and "LKR" from numeric text and fills invalid numbers with the median, trims and title-cases text, and removes empty rows, reporting each step to the user. The cleaned table is stored as CSV text in chunks of 5,000 lines in skyhigh_csv_chunks, with one record per upload in skyhigh_upload_history, all committed in one batch. Services load the latest upload of a type by reassembling its chunks. Figure 2.5 shows the collections.

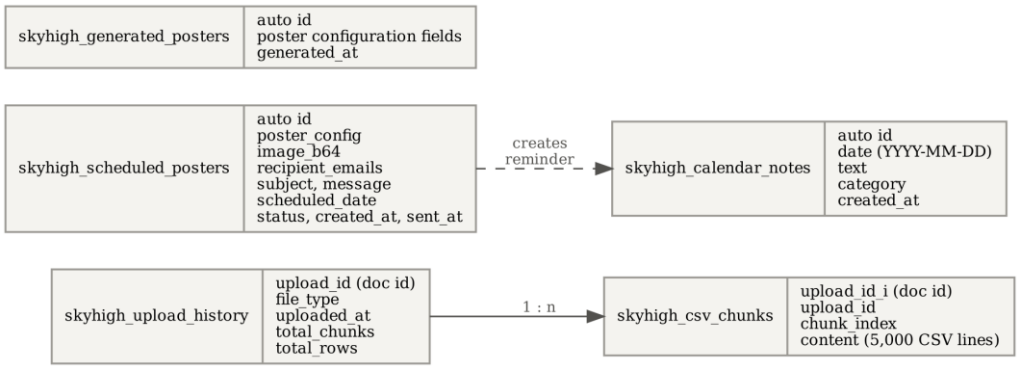


Figure 2.5: Firestore collections used by the marketing component

2.1.10 Redemption model and offer recommender design

The model service loads model.pkl and columns.json once and caches them. A request carries the customer profile, discount and three timing flags. Categorical values are checked against the saved encoders and unknown values are rejected with the valid options. The service builds six copies of the request, one per offer type, scores them with the random forest and returns the probabilities in descending order. The Recommender screen groups inputs into customer segment, behaviour, discount and timing (seasonal window: March–April, June–July, November, December; salary cycle: 20th to 1st; school holidays: April, August, December), re-scores automatically as inputs change, and lists the top five overstocked products in the selected inventory segment beside the result.

2.1.11 Customer intelligence design

From the latest customer upload the service computes the share of Gold and Platinum members, average spend and visits, counts by tier, age group and category, new versus returning customers (joined within 90 days of the latest join date, or at most one visit), disengagement risk (customers inactive longer than a chosen threshold, default 60 days, by tier) and a weighted RFM purchase-intent score:

$$intent = 0.4 \times R + 0.3 \times F + 0.3 \times M, \quad R = 100 \times (1 - days / max\ days),\ F = 100 \times visits / max\ visits,\ M = 100 \times spend / max\ spend$$

Scores of 66 and above are High, 33–65 Medium and below 33 Low; recency is weighted highest as the strongest short-term engagement signal [5]. The screen lists the ten highest-intent and highest-value customers.

2.1.12 Poster generation design

Figure 2.6 shows the poster pipeline. The form collects the target segment, offer and discount text, season, items, palette, style, brand, an optional tagline and theme, optional event details (date, time, location, valid-until date, branch) and a terms flag; starting from an inventory opportunity pre-fills product, category and brand. The prompt builder includes only the sections that apply, asks for a full-bleed design, and always reserves the bottom 12% as a plain footer band, plus a clear strip for event details or a disclaimer when needed. The Gemini 2.5 Flash Image model returns the artwork. Pillow then draws the event details as one line of white text on a dark band in the reserved strip, shrinking the font to fit, appends "Terms and Conditions Applied" if requested, and places the brand logo, scaled to 12% of the width, at the bottom centre. The generative model therefore contributes only the composition, while every date, place, condition and the logo are drawn exactly as entered (RQ3). Each generation's configuration, but not its image, is saved to history.

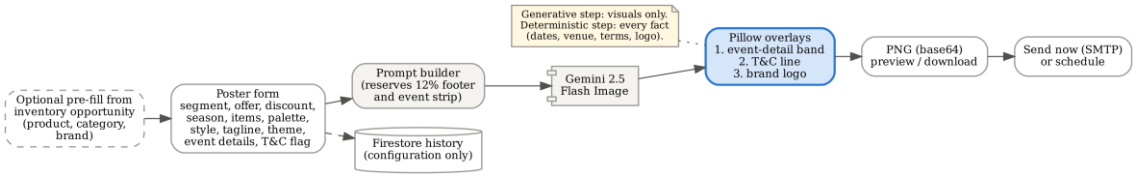


Figure 2.6: Poster generation pipeline

2.1.13 Delivery, scheduling and calendar design

E-mails are sent through Gmail's server using the simple mail transfer protocol (SMTP) on port 587, upgraded with transport layer security (TLS) and authenticated with an application password, one message per recipient with the poster attached; the service reports which recipients succeeded. A scheduled post stores the configuration, image, recipients, message and date with status pending, and adds a calendar note on that date. The user sends it from the scheduled list, after which it is marked sent with a timestamp, and re-sending is rejected (Figure 2.7). The Calendar tab shows notes by category (Reminder, Campaign Idea, Urgent, General) and a year heat-map of campaigns sent per day; the Hub shows a "Tomorrow" card.

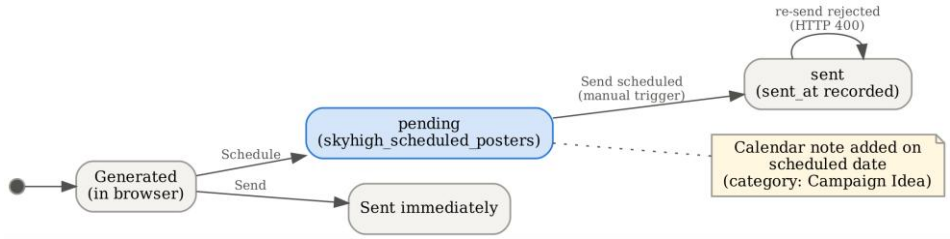


Figure 2.7: Lifecycle of a scheduled poster

2.1.14 KPI dashboard, reports and inventory link design

One function computes all KPIs for an optional date range: redemption rate, treatment-versus-control difference, redemption by offer type and the best offer, click-through by channel and the best channel, totals, monthly revenue from redeemed campaigns with the top offer each month, and the model's test accuracy. The Hub calls it for Today, This Month, Last Month, This Year, Last Year or a custom range, and the report service calls it for each selected year, month or quarter to build a PDF with ReportLab and matplotlib, so reports always agree with the dashboard (NFR2). The inventory service uses a second, separately named Firebase application for reads only. It provides the precomputed overstock alert filtered by category, the marketing_opportunities flagged PROMOTE with filters by category, gender, store and month, and a summary of items, excess units and value at risk (excess units × selling price).

2.1.15 Ethics, privacy, security and responsible AI

No real personal data was used, but the component was designed as if processing real data under Sri Lanka's Personal Data Protection Act (PDPA), No. 9 of 2022 [29]. Uploads lose any column whose name contains phone, email, dob, mobile, telephone, address, name or nic before storage; e-mails are sent to each recipient separately; and credentials are kept in environment variables and key files excluded from version control. The recommender shows all six scores rather than one instruction. Gender and age group are model inputs because they are standard segmentation variables in fashion retail, but their use should be reviewed for fairness before deployment. Posters are reviewed by the marketer, and the generative model is never trusted with prices,

dates or terms. Appendix B discloses the use of generative AI in producing this dissertation.

2.2 Commercialization Aspects of the Product

2.2.1 Value proposition and target users

The target users are marketing teams and store managers in small and mid-sized multi-branch fashion and lifestyle retailers in Sri Lanka and similar markets, who run regular promotions without data-science staff. The value proposition is one tool that indicates which offer each customer group is most likely to redeem and when, produces accurate posters in minutes, sends and schedules them, shows what worked, and points promotions at stock that needs to move. The expected benefits are fewer wasted discounts, faster campaign production and less overstock carried into the next season. These problems are costly at industry scale. IHL Group estimates that inventory distortion (the combined cost of out-of-stocks and overstocks) costs retailers worldwide about USD 1.7 trillion a year, equal to 6.2% of global retail sales [31], and McKinsey reports that retailers in the United States alone were holding about USD 740 billion in unsold goods, which they were clearing through steep markdowns that squeeze margins [32]. The expected benefits of the component are therefore fewer wasted discounts, faster campaign production and less overstock carried into the next season.

2.2.2 Market and competitive position

Existing tools fall into three price bands. At the low end, e-mail marketing services such as Mailchimp offer a free plan for up to 250 contacts and paid plans from about USD 13 a month (Essentials), USD 20 a month (Standard) and USD 350 a month (Premium); predictive segments are available only on the Standard and Premium plans [35]. Design tools with generative-AI features, such as Canva, are free for basic use, with Canva Pro costing about USD 144 a year per person [36]. At the high end, enterprise platforms such as Salesforce Marketing Cloud start at about USD 1,500 a month per organisation (Growth Edition) and USD 3,250 a month (Advanced Edition), billed annually [37], a cost well beyond most small Sri Lankan retailers.

Mid-market tools such as Klaviyo offer predictive analytics like customer lifetime value and churn risk [32], with pricing that scales with the number of customer profiles. The component sits between these bands: more analytical than a design or e-mail tool, far simpler and cheaper than an enterprise platform, and connected to inventory as part of the Smart Omni Retail Recommendation System. None of the tools reviewed ranks several offer types for an individual customer, keeps poster facts deterministic, or links promotions to overstock, which are the component's main differentiators (Table 1.2).

2.2.3 Business and sustainability model

A plausible model is a monthly subscription per retailer, priced by number of stores, as a module of the Smart Omni-Retail Recommendation System, with onboarding to import historical campaign data. Running costs are cloud hosting, Firestore usage and image-generation calls, which scale with the number of posters; volume e-mail would need a transactional e-mail provider.

2.2.4 Scalability, intellectual property and compliance

The stateless API and cached model scale horizontally. For many retailers, CSV chunks should move to a data warehouse or cloud object storage, scheduled sending to a managed scheduler, and the model must be retrained on each retailer's data with the leakage-safe feature computation as a required step. The code is the group's work under the student intellectual-property policies of the Sri Lanka Institute of Information Technology (SLIIT). Under the Gemini API Additional Terms of Service, Google does not claim ownership of generated content, but it may generate the same or similar content for others, and the user remains responsible for how generated content is used and shared [38]. Every image generated through the API also carries an invisible SynthID watermark identifying it as AI-generated [39]. A retailer adopting the component would therefore own and be accountable for its posters, but could not claim exclusive rights to the generated artwork itself, and should disclose AI use where advertising rules require it. The terms also require API users to be at least 18 and prohibit applications aimed at under-18s, which is consistent with a tool used by marketing staff. Real customer data would bring the component under the PDPA [29]:

a lawful basis for processing, purpose limitation, minimisation (already supported by identifier removal), secure storage and transfer, and the right to object to direct marketing, which requires an unsubscribe mechanism the prototype does not yet have.

2.2.5 Contribution to the Sustainable Development Goals

Table 2.7 relates the component to the Sustainable Development Goals (SDGs) [30]. The contribution is plausible rather than demonstrated.

Table 2.7: Relationship to the Sustainable Development Goals

Goal	Possible contribution	Possible negative effect
SDG 8: Decent Work and Economic Growth	Gives small retailers analytics usually limited to large firms	Automated design could reduce demand for some creative work
SDG 9: Industry, Innovation and Infrastructure	Demonstrates affordable AI adoption by local retail businesses	Dependence on foreign cloud and AI providers
SDG 12: Responsible Consumption and Production	Directs promotions at overstocked items, which may reduce unsold stock and waste	More effective promotion may also increase consumption

2.3 Testing and Implementation

2.3.1 Development environment and technology stack

Table 2.8 lists the technologies used.

Table 2.8: Technology stack

Layer	Technology	Use in the component
Frontend	Flutter (Dart), http, file_picker, fl_chart	Seven-tab application, charts, file upload
Backend	Python, FastAPI, Uvicorn, Pydantic	REST API and validation
Machine learning	scikit-learn, pandas, NumPy (XGBoost for comparison)	Random forest model, baselines and evaluation
Generative AI and imaging	Gemini 2.5 Flash Image (google-genai), Pillow	Poster artwork and deterministic overlays
Reporting	ReportLab, matplotlib	PDF reports with charts
Storage and e-mail	Firebase Firestore (firebase-admin), smtplib with Gmail	Data, history, schedules; delivery
Configuration and versioning	python-dotenv, Git and GitHub	Secrets from environment; branch IT22146588-Ruvindi

2.3.2 Component implementation

Upload and cleaning. The upload endpoint reads the multipart file with the encoding fallback, cleans it and saves it in chunks with a single batch commit, returning the row count, cleaning report, columns and a three-row preview. Deleting an upload removes its history record and all chunks in one batch.

Model training and serving. The training script loads the dataset, recomputes the past-redemption feature from earlier campaigns, label-encodes the categorical features, splits the data 80/20 with stratification and seed 42, trains the random forest (300 trees, maximum depth 10, minimum 50 samples per leaf, square-root features) and prints accuracy, ROC-AUC and feature importance. It saves the model with pickle and writes columns.json with the features, encoders, offer types, algorithm, hyperparameters, metrics, baseline rate, importance and the feature-computation method. The model service and the /model/info endpoint read these files, so the dashboard always shows the metrics of the model in use, and the API did not change when the algorithm did.

Analytics and reports. KPIs and customer intelligence operate on pandas data frames loaded from the latest uploads; the report endpoint returns the PDF as an attachment, with matplotlib in its non-interactive back end so charts render on a server.

Posters, e-mail and scheduling. The poster endpoint builds the prompt and overlay lines, calls Gemini, applies the overlays and returns the image as base64 for display and download; if Gemini returns no image, for example because a prompt was blocked by safety filters, an explanatory error is returned. The e-mail and scheduling endpoints decode the image, call the e-mail service or save the post, and add the calendar note. Table 2.9 summarises the backend modules.

Table 2.9: Backend modules and endpoints

Module	Responsibility	Main endpoints
upload_routes, cleaner_service, firebase_service	Upload, cleaning, storage, history	/upload, /upload-history, /upload/{id}/download
recommend_routes, model_service	Offer ranking, model metadata	/recommend, /model/info
customer_routes, customer_service	Segments, intent, at-risk	/customer-intelligence/
kpi_routes, report_routes, report_service	KPIs, periods, PDF reports	/kpis, /kpis/available-periods, /reports/generate
poster_routes, poster_service, email_service	Generation, history, e-mail, scheduling	/poster/generate, /poster/send-email, /poster/schedule, /poster/send-scheduled/{id}
calendar_routes	Campaign counts and notes	/calendar/campaigns, /calendar/notes
inventory_routes, inventory_service	Read-only inventory data	/inventory/overstock-suggestions, /inventory/marketing-opportunities

2.3.3 Integration with the other components

The marketing component integrates with the inventory component (Methsara W.M) through the shared team Firebase project (Figure 2.8). The inventory component publishes an overstock alert, listing overstocked products with store, brand, category, stock and reorder level, and marketing opportunities flagged PROMOTE with excess quantity and selling price. The marketing component shows the top overstocked products in the selected segment next to the recommended offer, presents the

opportunities with summary figures, category charts and filters in the Inventory Insights tab, and offers a Create Poster button that opens the Poster tab pre-filled with the product, category and brand. The link is read-only by design: it never writes to the shared project, uses a separate Firebase application object, and returns an error object that the screens display if the shared project is unavailable (TC24). No data is exchanged with the Smart Mirror or customer-tracking components in this version.

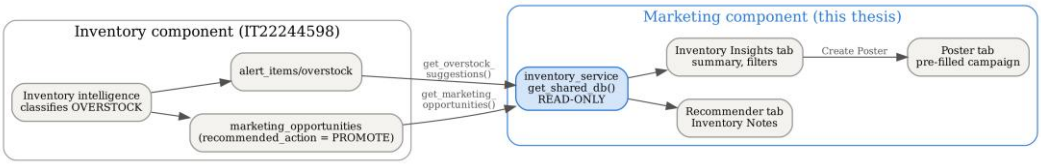


Figure 2.8: Integration with the inventory component

2.3.4 Verification and software testing

Twenty-five test cases were automated with the FastAPI test client, with Firestore replaced by the research dataset in memory so that the tests are repeatable without credentials, and with Gemini and the mail server called without credentials to test their failure paths; their success paths were verified manually through the application The cases cover the API (TC01), recommender (TC02–TC06), upload and cleaning (TC07–TC09), KPIs and reports (TC10–TC15), customer intelligence (TC16), posters (TC17–TC20), e-mail and scheduling (TC21–TC23), the inventory link (TC24) and the leakage fix (TC25). Because the dependency file does not pin pandas, the cleaning cases were run with pandas 3.0.5 and 2.3.3. Appendix A.2 gives every case with its expected and observed result.

Twenty-two cases passed and three defects were found (Table 2.10). D1 and D2 have one cause: the cleaner selects numeric columns by searching for keywords inside column names, and text columns by checking for the pandas object type. Under pandas 2.x the keyword "age" also matches age_group, which is turned into missing values, and identifiers such as CUST_001 are title-cased; under pandas 3.x text columns have a dedicated string type, so the numeric repair and title-casing are silently skipped. D3 arises because base64 decoding is not run in strict mode. The tests also showed that the redemption model responds sensibly to context, with the top probability for the

example customer falling from 50.7% to 38.6% when the seasonal window is switched off (TC06).

Table 2.10: Defects found and corrective actions

ID	Defect	Corrective action
D1	Keyword "age" matches age_group, which becomes missing; IDs are title-cased (pandas 2.x)	Match numeric columns by exact name or suffix; exclude _group and _id columns; never change the case of identifiers
D2	Text columns skipped under pandas 3.x because their type is no longer object	Use pandas.api.types.is_string_dtype; pin the pandas version
D3	Invalid base64 image data is not rejected	Decode with validate=True so the API returns HTTP 400

Three further observations are noted for improvement: invalid dates are filled with the previous row's date, which can invent dates; the KPI labelled "revenue uplift vs control" is actually a difference in redemption rates in percentage points; and the uplift KPI requires the treatment column to be coded 0/1, whereas the small demonstration file uses the words Treatment and Control.

2.3.5 Deployment, standards and reproducibility

The backend runs with Uvicorn from the backend directory, which holds the two Firebase key files and a .env file with the Gemini key and Gmail credentials; the Flutter application calls it at a configurable base URL, and retraining the model only requires running the training script and restarting the backend. The backend follows REST conventions, validates requests with Pydantic and distinguishes client errors (400, 404, 422) from upstream (502) and server (500) failures; business logic sits in services separate from routes, secrets stay out of source control, and data handling follows the PDPA principles of minimisation and purpose limitation [29]. The code is in the group repository (Methsara20/R26-IT-060), branch IT22146588-Ruvindi; running app/ml/train_model.py writes the model (model.pkl) and its metadata, encoders, hyperparameters and metrics (columns.json), all splits use seed 42, and Appendix A.3 lists software versions.

3 RESULTS AND DISCUSSION

3.1 Results

3.1.1 Redemption model performance

Table 3.1 reports all models on the same test set of 6,000 records after the leakage fix. The majority baseline is 55.9% accurate by predicting that everyone redeems and has no ranking ability (ROC-AUC 0.500); using each offer type's historical rate adds almost nothing (0.545). The three learned models with sensible settings perform almost identically: logistic regression 0.673, XGBoost 0.674 and the selected random forest 0.672, all within one cross-validation standard deviation (0.007) of each other. The random forest has the highest F1 (0.711) and recall (0.806) and equal-best accuracy (63.3%). The default random forest, with unlimited depth and single-record leaves, is clearly worse (0.651), which shows that limiting tree size was essential.

Table 3.1: Model performance on the held-out test set (corrected feature, n = 6,000)

Model	Accuracy	ROC-AUC	Precision	Recall	F1	Brier
Majority class	0.559	0.500	0.559	1.000	0.717	0.247
Offer-type historical rate	0.559	0.545	0.559	1.000	0.717	0.245
Logistic regression	0.632	0.673	0.642	0.772	0.701	0.224
XGBoost	0.631	0.674	0.648	0.742	0.692	0.224
Random forest (default)	0.622	0.651	0.641	0.736	0.685	0.231
Random forest (selected)	**0.633**	**0.672**	**0.636**	**0.806**	**0.711**	**0.225**

The result is stable (Table 3.2). Across five folds of the full dataset the selected model's ROC-AUC was 0.673 ± 0.007 and its accuracy 0.633 ± 0.006. Trained on 2018–2024 and tested on 2025 it reached 62.8% accuracy and a ROC-AUC of 0.666 against a 2025 base rate of 56.5%, so it generalises to a later year. Its confusion matrix on the test set is 1,096 true negatives, 1,549 false positives, 651 false negatives and 2,704 true positives; the model leans towards predicting redemption, the majority class, which does not affect recommendations because offers are ranked rather than thresholded.

Table 3.2: Robustness of the selected random forest

Evaluation	Accuracy	ROC-AUC	F1
Random 80/20 split	0.633	0.672	0.711
Five-fold cross-validation (mean ± s.d.)	0.633 ± 0.006	0.673 ± 0.007	0.711 ± 0.006
Chronological (train 2018–2024, test 2025)	0.628	0.666	0.708

3.1.2 Effect of the leakage fix

Table 3.3 and Figure 3.1 compare each model trained with the supplied and the corrected feature. With the supplied feature, XGBoost reached 73.5% accuracy and a ROC-AUC of 0.818 (the figures reported in the earlier version of this work), a deeper XGBoost configuration recorded 85.5% in the stored metadata, and the default random forest 0.723. After the fix all fell to about 0.65–0.67. Logistic regression was unaffected, because a linear model cannot exploit a signal that appears only within one customer, and the selected random forest changed little (0.678 to 0.672), because leaves of at least 50 records prevent it from isolating individual customers.

Table 3.3: Test ROC-AUC (accuracy) with the supplied and corrected feature

Model	Supplied feature (leaking)	Corrected feature
Logistic regression	0.672 (63.3%)	0.673 (63.2%)
XGBoost	0.818 (73.5%)	0.674 (63.1%)
Random forest (default)	0.723 (66.6%)	0.651 (62.2%)
Random forest (selected)	0.678 (63.8%)	0.672 (63.3%)

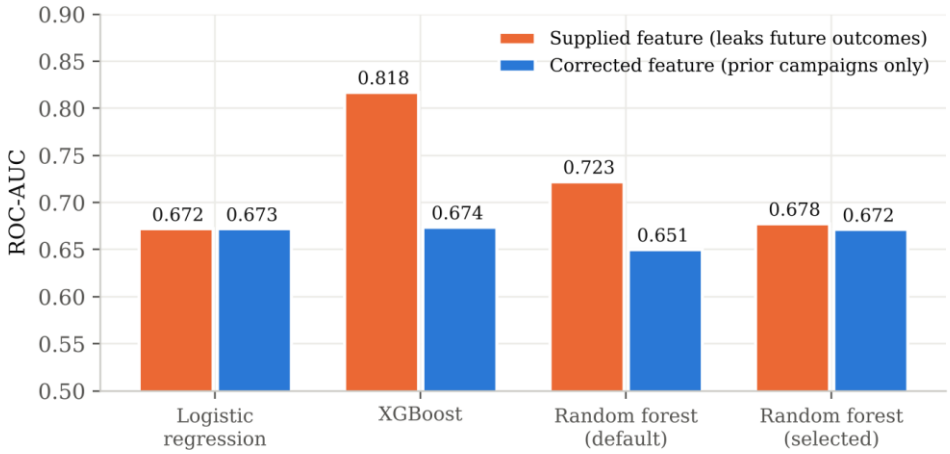


Figure 3.1: Test ROC-AUC with the supplied (leaking) and corrected feature

Feature-group ablation shows where the corrected model's signal comes from (Table 3.4). Removing the three timing flags lowers ROC-AUC from 0.672 to 0.600, the largest drop; removing offer type, discount and channel lowers it to 0.652; removing the corrected past-redemption rate changes nothing (0.673). The model's importance scores agree (Figure 3.2): the seasonal window (0.315) and salary cycle (0.127) dominate, followed by recency, visit frequency, offer type and discount.

Table 3.4: Feature-group ablation of the selected random forest (test set)

Inputs used	**Accuracy**	**ROC-AUC**
All fifteen features	0.633	0.672
Without timing flags	0.584	0.600
Without offer type, discount and channel	0.617	0.652
Without past redemption rate	0.633	0.673
Timing, offer, discount and channel only	0.632	0.658

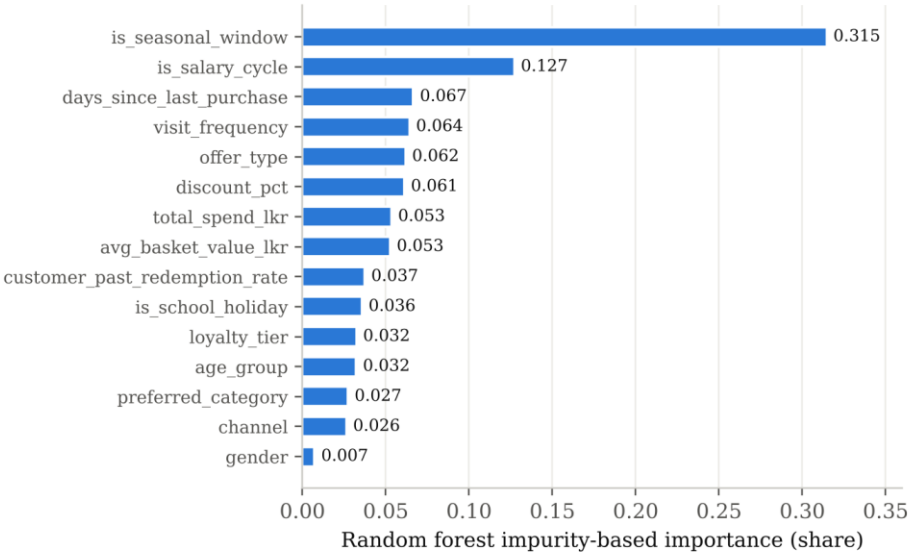


Figure 3.2: Feature importance of the selected random forest

3.1.3 Calibration and targeting value

Figure 3.3 plots the observed redemption rate within each decile of predicted probability. The points follow the diagonal through the middle of the range, where most predictions lie; at the extremes the forest is slightly cautious, predicting 34.9% in the lowest decile where 26.5% redeemed and 74.2% in the highest where 79.5% did. For targeting, the 10% of test campaigns with the highest scores were redeemed at 79.5%, a lift of 1.42 over the 55.9% average, against 26.5% in the lowest 10%.

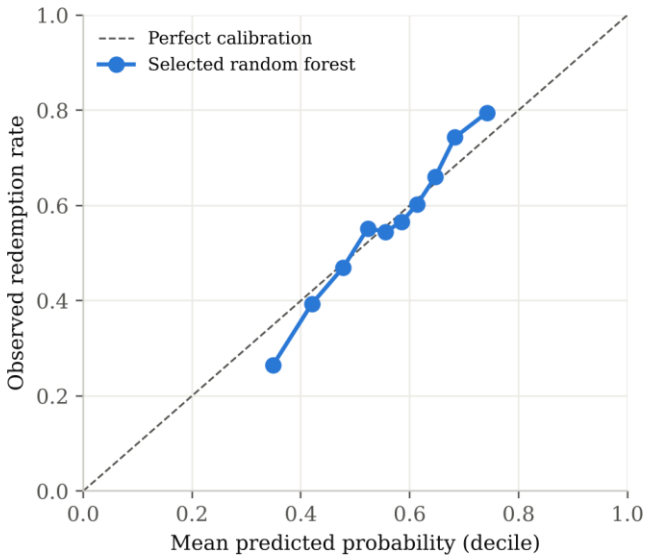


Figure 3.3: Calibration of predicted redemption probabilities (test set, deciles)

3.1.4 Offer ranking

For RQ2, every test record was re-scored under all six offer types. In 1,040 records the offer actually sent was the model's top-ranked offer, and these were redeemed at 61.6%; the other 4,960 were redeemed at 54.7%. The model's top offer was Flash Sale for 2,374 records, BOGO for 2,332, Discount for 919, Loyalty Reward for 286, Seasonal Sale for 74 and New Arrival for 15; the mean gap between a customer's best and worst offer was 6.2 percentage points, smaller than the effect of timing. Table 3.5 shows an example: a woman aged 35–45 in the Silver tier, preferring coats and app notifications, 116 days since her last purchase, a 10% discount, inside a seasonal window. Flash Sale ranks first at 50.7%; outside the seasonal window the top probability falls to 38.6% (TC06). For her actual offer (New Arrival), raising the discount from 10% to 30% raises the probability from 47.5% to 51.7%, with no further gain beyond 30%.

Table 3.5: Example recommendation for one test customer

Offer type	Predicted redemption probability
Flash Sale	50.7%
BOGO	50.5%
Discount	50.2%
Loyalty Reward	47.7%
New Arrival	47.5%
Seasonal Sale	46.5%

3.1.5 Product, integration and user validation results

All functional requirements are implemented. On the research dataset the dashboard reports a 55.9% redemption rate, Flash Sale as the best offer (61.7%), App Push as the best channel (69.2% click-through), a treatment-versus-control difference of 10.8 percentage points and LKR 178.9 million from redeemed campaigns, matching an independent calculation (TC10). Customer intelligence identifies 214 high-, 1,654 medium- and 632 low-intent customers, and 1,700 customers (68.0%) inactive for more than 60 days. The read-only inventory link shows overstock suggestions in the Recommender, lists opportunities with filters, and pre-fills the poster form, and the screens continue to work when the shared project is unavailable (TC24).

No structured user evaluation had been completed at the time of this draft, and none is reported; the evidence still to be added, in line with the PP3 requirement for user feedback, is listed in Appendix A.4 (Table A.3).

3.2 Research Findings

The earlier accuracy of the redemption model was inflated by target leakage. After correction, a regularised random forest predicts redemption with a modest but consistent improvement over baselines (ROC-AUC 0.672 against 0.500 and 0.545), generalises to a later year, is barely affected by the leak, and produces sensible probabilities. Timing context is the strongest predictor and offer type matters less. Offers ranked first by the model were redeemed more often (61.6% against 54.7%). The poster pipeline keeps all facts deterministic, and the component meets its

functional requirements with three defects identified. Table 3.6 summarises the research contributions and the strength of the evidence for each.

Table 3.6: Research contributions and strength of evidence

Contribution	Description	Evidence
Leakage audit and correction	Identification of a leave-one-out customer-history feature that leaked current and future outcomes, its correction, and its measured effect across models	Strong on this dataset (Table 3.3, TC25)
Offer ranking by conditional redemption probability	A regularised random forest with offer type as input, used to rank six offers per customer	Moderate: offline alignment, synthetic data
Importance of timing context	Timing flags carry most of the signal	Moderate: ablation on synthetic data
Deterministic-fact poster pipeline	Generated artwork with reserved bands and code-drawn facts and branding	Functional tests; user evaluation pending

3.3 Discussion

3.3.1 Interpretation of findings

The most important finding is methodological. A model that appeared to reach 73.5% accuracy, and up to 85.5% in a deeper configuration, was in fact a 63% model; the difference came from one feature whose convenient definition contained the answer. The corrected result is what a retailer would experience when using past campaigns to plan future ones, and the chronological test on 2025 supports it.

A ROC-AUC of 0.672 means that for a random redeemed and non-redeemed campaign, the model scores the redeemed one higher about two times in three. That is moderate: enough to rank and prioritise, as the top-decile lift of 1.42 and the seven-point advantage of top-ranked offers show, but not enough to treat any single prediction as certain. This is consistent with the problem, since whether a customer redeems depends on much that is not in the data.

The ablation shows that in this dataset when a campaign is sent matters more than which offer is sent or to whom. Seasonal windows and salary-cycle timing carry most of the signal, which is plausible for Sri Lankan retail with its festival peaks and salary-date spending, but it reflects how the synthetic data was generated and must be

confirmed on real data. Practically, the timing switches are among the most important inputs, so future versions should derive them automatically from the campaign date.

Because logistic regression, XGBoost and the random forest perform equally well after the fix, the remaining relationships appear largely additive. The random forest was selected because it matched the others on ranking, gave the best F1 and recall, was the least sensitive to leakage, needs only a few intuitive settings and trains in parallel. A logistic model would be a defensible, more interpretable alternative.

3.3.2 Comparison with literature and existing solutions

The leakage finding matches the pattern described by Kaufman et al. and by Kapoor and Narayanan [19], [20]: a feature built with information from the evaluation period, producing over-optimistic results that disappear under correct evaluation, with flexible models exploiting it most. Reporting results against majority and offer-rate baselines follows the principle that performance should be judged against what a practitioner could achieve without a model. The mild compression of the forest's probabilities at the extremes is consistent with the behaviour of averaged trees reported by Niculescu-Mizil and Caruana [18], and errs on the side of not overstating confidence.

Compared with uplift modelling [8], [10], [9], the recommender answers a narrower question: it ranks offers by redemption probability, not by the incremental effect of sending them. Because offers were assigned close to randomly, the comparison between offers is reasonably fair, but redemption probability alone could favour customers who would buy anyway; the dashboard's treatment-versus-control difference (10.8 points) provides the campaign-level view. Compared with the approaches in Table 1.2, the component's distinguishing features are offer ranking per customer, generated artwork with deterministic facts [24], [25], a closed loop from delivery to reporting, and the connection to inventory surplus.

3.3.3 Limitations and threats to validity

Table 3.7 lists the main limitations.

Table 3.7: Limitations and their effects

Limitation	Effect	Section
Synthetic data	Results show recovery of the generator's patterns, not real-world accuracy	2.1.2
Response model, not uplift	May favour customers who would redeem anyway; no causal claim	1.1.6
Observational ranking check	61.6% vs 54.7% is not a controlled test	3.1.4
Fixed customer attributes	Spend, visits and recency describe each customer at one unknown time	2.1.2
Gender and age as inputs	Risk of unfair targeting without review	2.1.15
Generative model dependence	Quality, safety filtering, cost and availability depend on an external API	2.1.12
Cleaning defects D1–D3; manual scheduled sending	Data may be altered depending on pandas version; posts not sent automatically	2.3.4
No user validation yet	Usability and usefulness not yet evidenced	3.1.5

3.3.4 Implications

For practitioners, customer-history features must be computed as of the prediction date, and a large gain from a flexible model over a linear one should prompt a leakage check before it is reported. For the retailer, the results suggest prioritising campaign timing and using model scores to rank offers and audiences rather than to make yes-or-no decisions. For generative-AI tools in business, the poster pipeline shows a simple way to keep the model away from the facts the business must stand behind.

3.4 Summary of Each Student's Contribution

Table 3.8 summarises the division of work in the group; each member's own dissertation reports their component in detail.

Table 3.8: Summary of each student's contribution

Member	Student ID	Component	Main contribution
Methsara W.M	IT22244598	AI-Driven Smart Inventory and Stock-Flow Optimization	Inventory component and the shared team Firebase project, including the overstock alerts and promotion opportunities used by the marketing component
Wijetunga J A R K	IT22146588	AI-Driven Personalized Marketing Intelligence	RFM features and Random Forest to predict which customers are likely to redeem a promotion, then generate promotional posters for those customers, promoting only products that are in stock.
T A Silva	IT22284952	AI-Based Virtual Try-On and Size Recommendation (Smart Mirror)	Predicts body measurements and brand-specific sizes with Random Forest models, shows the garment on the customer's own photo using Vertex AI, and suggests in-stock items through a RAG-based AI stylist.
L A D A S Perera	IT22243812	GPS-Based Customer Behaviour Tracking with Staff Assistance	Classifies in-store movement as browsing or transiting from phone GPS data and alerts staff when a customer seems to need help, without any beacon or camera hardware.

My own contribution comprises the synthetic campaign dataset; the upload and cleaning pipeline with chunked Firestore storage; customer intelligence; the redemption model and offer recommender, including the leakage audit, the comparison of models and the tuning of the random forest; the poster pipeline with deterministic overlays; e-mail delivery, scheduling and the promotion calendar; the KPI dashboard and PDF reports; the read-only integration with the inventory component; the Flutter screens of the marketing component; and the verification tests. In product terms this is a working tool covering the cycle from data upload to reporting, with seven screens and 29 API endpoints. The shared team Firebase project and its inventory collections were created by the inventory component, which my component only reads.

4 CONCLUSIONS AND RECOMMENDATIONS

4.1 Conclusions

This dissertation designed, implemented and evaluated an AI-driven marketing intelligence component for a multi-channel fashion retailer. It cleans and stores campaign data without personal identifiers, describes customers through segments, purchase intent and disengagement risk, ranks six offer types for any customer profile, generates posters whose facts are always exact, delivers and schedules campaigns, reports performance consistently on screen and in PDF, and links promotion planning to the inventory component's overstock data.

For RQ1, once a leaking feature was corrected, a regularised random forest predicted redemption with 63.3% accuracy, a ROC-AUC of 0.672 and an F1 of 0.711, clearly above majority (0.500) and offer-rate (0.545) baselines, stable across folds and on a later year, and on a par with logistic regression and XGBoost. For RQ2, offers ranked first by the model were redeemed at 61.6% against 54.7% for other offers in an offline comparison. For RQ3, separating generated artwork from deterministically drawn facts keeps dates, venues, terms and the logo exact by construction. For RQ4, a separate read-only connection allowed inventory surplus data to be used in planning without risk to the inventory component. These findings rest on synthetic data and await validation with users and real campaigns.

4.2 Achievement of Objectives

Table 4.1 relates each specific objective to the evidence.

Table 4.1: Objective-achievement matrix

ID	Objective (short form)	Status	Evidence and remaining gap
O1	Dataset and cleaning pipeline	Achieved, with defects	29,996-record dataset; identifier removal tested (TC07); D1–D2 to fix
O2	Leakage-free redemption model	Achieved	Section 3.1, TC25; real-data validation outstanding
O3	Offer-ranking recommender	Achieved	TC03–TC06; Section 3.1.4
O4	Customer intelligence	Achieved	TC16; Section 3.1.5
O5	Posters with deterministic facts	Achieved	TC17–TC20; poster-quality evaluation outstanding
O6	Delivery, scheduling, calendar	Achieved	TC21–TC23; automatic sending outstanding; D3 to fix
O7	KPI dashboard and PDF reports	Achieved	TC10–TC15
O8	Inventory integration and verification	Achieved; user validation pending	TC24; Section 3.1.5

4.3 Recommendations and Future Work

Table 4.2 lists the recommended next steps.

Table 4.2: Recommendations and future work

Recommendation	Purpose
Validate with real campaign data and users	Measure real accuracy, usability and trust; complete the evidence in Table A.3
Build a multi-treatment uplift model	Recommend the offer with the largest incremental effect rather than the highest redemption probability [9], [10]
Derive timing flags from the campaign date	Remove manual input of the most influential features
Add per-recommendation explanations	Show SHAP contributions for each ranked offer [15]
Review demographic inputs for fairness	Check whether gender and age lead to unequal treatment
Automate sending, add unsubscribe handling and fix defects	Reliable, compliant delivery; resolve D1–D3 and pin dependencies

REFERENCE LIST

[1] M. Wedel and P. K. Kannan, "Marketing analytics for data-rich environments," J. Marketing, vol. 80, no. 6, pp. 97–121, Nov. 2016, doi: 10.1509/jm.15.0413.

[2] T. Davenport, A. Guha, D. Grewal, and T. Bressgott, "How artificial intelligence will change the future of marketing," J. Acad. Marketing Sci., vol. 48, no. 1, pp. 24–42, 2020, doi: 10.1007/s11747-019-00696-0.

[3] M.-H. Huang and R. T. Rust, "A strategic framework for artificial intelligence in marketing," J. Acad. Marketing Sci., vol. 49, no. 1, pp. 30–50, 2021, doi: 10.1007/s11747-020-00749-9.

[4] P. Chandon, B. Wansink, and G. Laurent, "A benefit congruency framework of sales promotion effectiveness," J. Marketing, vol. 64, no. 4, pp. 65–81, Oct. 2000, doi: 10.1509/jmkg.64.4.65.18071.

[5] P. S. Fader, B. G. S. Hardie, and K. L. Lee, "RFM and CLV: Using iso-value curves for customer base analysis," J. Marketing Res., vol. 42, no. 4, pp. 415–430, Nov. 2005, doi: 10.1509/jmkr.2005.42.4.415.

[6] W. Verbeke, K. Dejaeger, D. Martens, J. Hur, and B. Baesens, "New insights into churn prediction in the telecommunication sector: A profit driven data mining approach," Eur. J. Oper. Res., vol. 218, no. 1, pp. 211–229, 2012, doi: 10.1016/j.ejor.2011.09.031.

[7] E. Ascarza, "Retention futility: Targeting high-risk customers might be ineffective," J. Marketing Res., vol. 55, no. 1, pp. 80–98, 2018, doi: 10.1509/jmr.16.0163.

[8] P. Gutierrez and J.-Y. Gérardy, "Causal inference and uplift modelling: A review of the literature," in Proc. 3rd Int. Conf. Predictive Applications and APIs, Proc. Mach. Learn. Res., vol. 67, 2017, pp. 1–13.

[9] P. Rzepakowski and S. Jaroszewicz, "Decision trees for uplift modeling with single and multiple treatments," Knowl. Inf. Syst., vol. 32, no. 2, pp. 303–327, 2012, doi: 10.1007/s10115-011-0434-0.

[10] S. R. Künzel, J. S. Sekhon, P. J. Bickel, and B. Yu, "Metalearners for estimating heterogeneous treatment effects using machine learning," Proc. Natl. Acad. Sci. USA, vol. 116, no. 10, pp. 4156–4165, 2019, doi: 10.1073/pnas.1804597116.

[11] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?," in Adv. Neural Inf. Process. Syst. 35 (NeurIPS 2022), Datasets and Benchmarks Track, 2022.

[12] L. Breiman, "Random forests," Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[13] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining (KDD '16), San Francisco, CA, USA, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[14] D. W. Hosmer, S. Lemeshow, and R. X. Sturdivant, Applied Logistic Regression, 3rd ed. Hoboken, NJ, USA: Wiley, 2013.

[15] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Adv. Neural Inf. Process. Syst. 30, 2017, pp. 4765–4774.

[16] A. Barredo Arrieta et al., "Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," Inf. Fusion, vol. 58, pp. 82–115, Jun. 2020, doi: 10.1016/j.inffus.2019.12.012.

[17] T. Fawcett, "An introduction to ROC analysis," Pattern Recognit. Lett., vol. 27, no. 8, pp. 861–874, 2006, doi: 10.1016/j.patrec.2005.10.010.

[18] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in Proc. 22nd Int. Conf. Mach. Learn. (ICML), Bonn, Germany, 2005, pp. 625–632.

[19] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," ACM Trans. Knowl. Discov. Data, vol. 6, no. 4, Art. no. 15, Dec. 2012, doi: 10.1145/2382577.2382579.

[20] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, Art. no. 100804, Sep. 2023, doi: 10.1016/j.patter.2023.100804.

[21] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in Adv. Neural Inf. Process. Syst. 33, 2020, pp. 6840–6851.

[22] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 10684–10695.

[23] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, "Hierarchical text-conditional image generation with CLIP latents," arXiv:2204.06125, 2022.

[24] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," in Adv. Neural Inf. Process. Syst. 35, 2022, pp. 36479–36494.

[25] J. Chen, Y. Huang, T. Lv, L. Cui, Q. Chen, and F. Wei, "TextDiffuser: Diffusion models as text painters," in Adv. Neural Inf. Process. Syst. 36, 2023.

[26] Google, “Nano Banana image generation,” Gemini API documentation. [Online]. Available: https://ai.google.dev/gemini-api/docs/image-generation (accessed Oct. 10, 2026). [27] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825–2830, 2011.

[28] Google, “Usage and limits,” *Firestore Documentation*, Firebase. [Online]. Available: https://firebase.google.com/docs/firestore/quotas

[29] Parliament of the Democratic Socialist Republic of Sri Lanka, "Personal Data Protection Act, No. 9 of 2022," Colombo, Sri Lanka, 2022.

[30] United Nations General Assembly, "Transforming our world: The 2030 Agenda for Sustainable Development," Resolution A/RES/70/1, Sep. 2015.

[31] Mailchimp, “Audience management tools,” Mailchimp features. [Online]. Available: https://mailchimp.com/features/personalization/ (accessed Oct. 10, 2026).

[32] Klaviyo, “Understanding Klaviyo’s predictive analytics,” Klaviyo Help Center. [Online]. Available: https://help.klaviyo.com/hc/en-us/articles/360020919731 (accessed Oct. 10, 2026).

[33] Salesforce, “Engagement scoring benefits & insights,” Einstein Engagement Scoring in Marketing Cloud, Trailhead. [Online]. Available: https://trailhead.salesforce.com/content/learn/modules/einstein-engagement-scoring-in-marketing-cloud/explore-engagement-scoring- (accessed Oct. 10, 2026).

[34] Canva, “Free AI image generator: Text to image,” Canva. [Online]. Available: https://www.canva.com/ai-image-generator/ (accessed Oct. 10, 2026).

[35] Mailchimp, “Marketing plans and pricing,” Mailchimp. [Online]. Available: https://mailchimp.com/pricing/marketing/ (accessed Oct. 10, 2026).

[36] Canva, “Canva pricing,” Canva. [Online]. Available: https://www.canva.com/pricing/ (accessed Oct. 10, 2026).

[37] Salesforce, “Marketing Cloud pricing,” Salesforce. [Online]. Available: https://www.salesforce.com/marketing/pricing/ (accessed Oct. 10, 2026).

[38] Google, “Gemini API additional terms of service,” Google AI for Developers. [Online]. Available: https://ai.google.dev/gemini-api/terms (accessed Oct. 10, 2026).

[39] Google, “Nano Banana image generation,” Gemini API documentation. [Online]. Available: https://ai.google.dev/gemini-api/docs/image-generation (accessed Oct. 10, 2026).

GLOSSARY

Calibration – Agreement between predicted probabilities and observed frequencies.

Data leakage – Use of information during training or evaluation that would not be available when the model is used, making results over-optimistic.

Deterministic overlay – Text or graphics drawn onto a generated image by code, exactly as specified.

Purchase-intent score – In this dissertation, a weighted RFM score from 0 to 100.

Random forest – An ensemble of decision trees, each trained on a bootstrap sample with random feature subsets, whose predictions are averaged.

Redemption – A customer using the offer they received.

Uplift – The change in the probability of an outcome caused by a treatment such as an offer.

APPENDICES

Appendix A: Extended Results and Research Instruments

A.1 Leakage fix in the training script

The function below was added to app/ml/train_model.py and is called from build_training_frame() before encoding.

```
def recompute_past_redemption_rate(df):
    df = df.copy()
    df["_order"] = np.arange(len(df))
    df = df.sort_values(["customer_id", "sent_date", "campaign_id"])
    prior_redeemed = df.groupby("customer_id")[TARGET].cumsum() - df[TARGET]
    prior_count = df.groupby("customer_id").cumcount()
    df["customer_past_redemption_rate"] = np.where(
        prior_count > 0, prior_redeemed / prior_count.replace(0, 1),
PRIOR_RATE
    ).round(4)
    return df.sort_values("_order").drop(columns="_order")
```

A.2 Full test-case record

Table A.1 gives every automated test case with its expected and observed result.

Table A.1: Detailed test-case record

ID	Scenario	Expected result	Observed result	Result
TC01	Health check endpoint	200 and status ok	200 {'status': 'ok'}	Pass
TC02	Model metadata served from columns.json	Corrected metrics (0.633, 0.672)	accuracy 0.6333, ROC-AUC 0.6718, rows 29996	Pass
TC03	Rank offers for a valid customer profile	Six offers, sorted descending, top = first	6 offers ranked; top Flash Sale (0.5074)	Pass
TC04	Unknown category value	HTTP 400 listing valid options	400: 'Laptop' is not a recognized value for 'preferred_category'....	Pass
TC05	Required field missing	HTTP 422 from schema validation	422 (validation error)	Pass
TC06	Timing context changes the score	Seasonal window raises predicted probability	top probability 0.3864 outside vs 0.5074 inside seasonal window	Pass
TC07	Customer CSV containing a name column	PII column removed, 500 rows saved	pandas 3.0.5: name column removed, 500 rows saved. pandas 2.3.3: age_group converted to missing values and response failed (D1)	Pass / Fail
TC08	Duplicates, e-mail column, '1,200 LKR' text, bad value and bad date	Duplicate dropped, PII dropped, numbers coerced, bad values repaired	pandas 2.3.3: duplicate removed, e-mail dropped, '1,200 LKR' → 1200, invalid value filled with median, invalid date repaired. pandas 3.0.5: numeric repair skipped (D2)	Fail / Pass
TC09	Unsupported file_type	HTTP 400	400	Pass
TC10	All-time KPIs on the research dataset	Match independent pandas computation	redemption 55.9%, best offer Flash Sale (61.7%), best channel App Push (69.2%), treatment-control 10.8 pp	Pass
TC11	Date-range filter for 2025	4,610 campaigns	total_campaigns 4610	Pass

ID	Scenario	Expected result	Observed result	Result
TC12	Range with no data	no_data_in_range flag, no crash	{"no_data_in_range": true, "total_campaigns": 0, "total_customers": 2500}	Pass
TC13	Available periods derived from data	2018-2025 only	8 years, 96 months, 32 quarters	Pass
TC14	PDF for two quarters	Valid PDF returned	200, 189,851 bytes PDF	Pass
TC15	Invalid period type	HTTP 400	400	Pass
TC16	Segments, RFM intent and at-risk on 2,500 customers	All sections returned	2500 customers; intent {'Medium': 1654, 'Low': 632, 'High': 214}; at risk 1700 (68.0%)	Pass
TC17	Prompt builder conditional sections	Prompt reflects only the options chosen	Offer sentence omitted for 'Not Applicable'; event strip and disclaimer space reserved when requested	Pass
TC18	Event-detail overlay on a 1024x1024 image	Text band confined to reserved area	Band drawn only in the reserved strip; rest of image unchanged	Pass
TC19	Brand logo watermark	Logo placed bottom-centre in the footer band	logo pixels in rows 900-996 (bottom band), cols 464-560	Pass
TC20	Generation with no Gemini key configured	Clear HTTP 502, no crash	502: Image generation failed: GEMINI_API_KEY not set in .env	Pass
TC21	Send with no SMTP credentials	HTTP 502 with explanatory message	502: Email credentials not configured on the server.	Pass
TC22	Corrupted image data	HTTP 400	HTTP 502 'Email credentials not configured': invalid base64 was accepted and the request reached the SMTP stage (D3)	Fail
TC23	Send a scheduled post that does not exist	HTTP 404	404: Scheduled post not found.	Pass
TC24	Inventory data when shared project unavailable	Graceful error object, no exception	{"error": "Shared Firebase not connected"}	Pass
TC25	Leakage regression test on training frame	No row uses its own or later outcomes	Feature equals mean of strictly earlier outcomes for all 29,996 rows; first campaign = 0.5	Pass

A.3 Model and environment details

Table A.2 lists the configuration of the selected model and the evaluation environment.

Table A.2: Model configuration and environment

Item	Value
Model	scikit-learn RandomForestClassifier: 300 trees, max_depth 10, min_samples_leaf 50, max_features sqrt, random_state 42
Tuning	Five-fold stratified grid search on the training set: max_depth {8, 10, 12, 16, None} × min_samples_leaf {5, 20, 50} × max_features {sqrt, 0.5}; best cross-validated ROC-AUC 0.671
Inputs	6 label-encoded categorical and 9 numeric features (Section 2.1.3)
Split and validation	80/20 stratified, seed 42 (23,996 / 6,000); five-fold cross-validation on the full data; chronological 2018–2024 / 2025 (25,386 / 4,610)
Evaluation environment	Python 3.13.16, scikit-learn 1.9.1, XGBoost 3.4.1, pandas 3.0.5 (2.3.3 for cleaning checks), FastAPI 0.143.0, Pillow 12.3.0, ReportLab 5.0.1

Appendix B: Academic Integrity and AI Use Disclosure

This appendix records how generative AI tools were used in producing this dissertation, in line with the module policy that AI may assist the work but cannot become its owner. AI used as part of the product (the Gemini image model) is described in Chapter 2; this disclosure, summarised in Table B.1, covers AI used as an assistant.

Table B.1: AI use disclosure

AI tool	Purpose	Representative workflow	Human verification performed	Sections affected
Claude (Anthropic)	Drafting text; re-running and tuning the model evaluation; identifying the leakage; producing diagrams, charts and the test suite			